

Understanding the LaTeX in main.tex

A beginner’s walkthrough of every command used in the NanoFlow report

This guide explains the LaTeX “syntax” (the commands) that appear in your `main.tex` file, grouped by topic. For each one you get: what it looks like in your file, and what it actually does. Nothing here changes your paper — it is just a reference so you can explain the file confidently.

A quick mental model first: **a LaTeX command almost always starts with a backslash (\)**, like `\section` or `\textbf`. Many commands take an *argument* in curly braces `{ }`, and some take *options* in square brackets `[ ]`. So `\command[option]{argument}` is the general shape. Anything after a percent sign `%` on a line is a **comment** — LaTeX ignores it, so it is just a note to yourself.

Contents

1. How the whole document is put together

Every LaTeX file has two parts:

1. The **preamble** — everything from the top of the file down to `\begin{document}`. This is *setup*: which document style to use, which add-on packages to load, and any custom shortcuts. Nothing here prints directly.
2. The **body** — everything between `\begin{document}` and `\end{document}`. This is the actual content that becomes pages.

```
\documentclass[manuscript,review]{acmart}    % preamble starts here
...packages and settings...
\begin{document}
...your paper...
\end{document}
```

Environments. Whenever you see a matching pair `\begin{something} ... \end{something}`, that is called an *environment*. The `document` environment above is the biggest one; later you will see `abstract`, `figure`, `table`, `tabular`, `itemize`, `equation`, and `algorithm` environments.

How a PDF is produced. The `.tex` file is *compiled* into a PDF. Because the paper has citations and cross-references, the tool has to run a few times: LaTeX runs once, then `bibtex` builds the reference list from the `.bib` file, then LaTeX runs a couple more times so all the “Figure 3” and “[12]” numbers settle. The `latexmk` tool just does all of that for you in one command.

2. The document class and packages

2.1 \documentclass

```
\documentclass[manuscript,review]{acmart}
```

This is always the very first line. It chooses the overall *style* of the document. Here the class is `acmart` (the official style of the ACM, the Association for Computing Machinery — the publisher your source paper uses). The words in brackets, `manuscript` and `review`, are *options* that tell `acmart` to use the single-column, larger-font “manuscript” layout with “review” spacing. Your assignment specifically required this exact line.

2.2 `\usepackage` — loading add-ons

LaTeX by itself is fairly basic; **packages** add features. You load one with `\usepackage{name}`. Here is every package in your file and why it is there:

```
\usepackage{graphicx}      % lets you insert images (\includegraphics)
\usepackage{booktabs}      % nicer horizontal lines in tables
\usepackage{amsmath}       % proper math typesetting (equations, \frac, ...)
\usepackage{amssymb}       % extra math symbols
\usepackage{xspace}        % smart spacing after your custom \nf command
\usepackage{multirow}      % lets a table cell span several rows
\usepackage{algorithm}     % the floating "Algorithm 1" box
\usepackage{algpseudocode} % the pseudocode commands (\State, \For, ...)
\usepackage{tikz}          % a drawing package (loaded, available if needed)
```

The line right after loads a `tikz` “library” (an add-on for the drawing package):

```
\usetikzlibrary{shapes.geometric, arrows.meta, positioning, calc, fit, backgrounds}
```

2.3 `\graphicspath`

```
\graphicspath{{figures/}}
```

This tells LaTeX “look inside the `figures/` folder for images.” That is why later you can write `\includegraphics{fig1_transformer.png}` without typing the folder name every time.

3. Preamble settings and custom shortcuts

3.1 Turning off ACM boilerplate

Because your paper is a class reproduction, not a real ACM submission, a few lines switch off the copyright notice and reference block that `acmart` would normally force:

```
\settopmatter{printacmref=false}      % no "ACM Reference Format" block
\renewcommand\footnotetextcopyrightpermission[1]{} % blank out the copyright footnote
\pagestyle{plain}                      % simple page numbers
\setcopyright{none}                    % no copyright statement
\acmConference[Reproduction Report]{CSE 200: ...}{...}{...}
\acmYear{2026}
\copyrightyear{2026}
```

The `\acmConference`, `\acmYear`, and `\copyrightyear` lines just fill in the venue/date fields that `acmart` expects.

3.2 `\newcommand` — making your own shortcut

```
\newcommand{\nf}{\textsc{NanoFlow}\xspace}
```

This *defines a new command*. After this line, every time you type `\nf` in the paper, LaTeX replaces it with `\textsc{NanoFlow}` (which prints “NANOFLOW” in small capitals). This is handy because the name appears dozens of times — if you ever wanted to restyle it, you would change one line instead of hundreds. The `\xspace` at the end politely adds a space after the word when one is needed (so “runs” does not become “NanoFlowruns”) but not before punctuation.

The next three define math shortcuts the same way:

```
\newcommand{\Bdense}{B_{\mathit{Dense}}} % prints B with subscript "Dense"
\newcommand{\Pmodel}{P_{\mathit{Model}}}
\newcommand{\Dmodel}{D_{\mathit{model}}}
```

Here `_` means “subscript” and `\mathit{...}` makes the subscript word italic. So `\Bdense` is a shortcut for the symbol B_{Dense} .

3.3 `\renewcommand` and `\let`

`\newcommand` makes a *new* command; `\renewcommand` *changes an existing* one. You use it here to set the short author name shown in page headers:

```
\renewcommand{\shortauthors}{Zhu et al.}
```

`\let` is a lower-level “copy this meaning” command. We used it once as a fix (explained in Section ??).

4. The title block (authors, abstract, title)

Everything that makes the “header” of the paper sits between `\begin{document}` and `\maketitle`.

```
\title{NanoFlow: Towards Optimal Large Language Model Serving Throughput}

\author{Kan Zhu}
\affiliation{%
  \institution{University of Washington}
  \country{USA}
}
```

- `\title{...}` sets the paper title.
- `\author{...}` names one author. You repeat this block for each of the 17 authors.
- `\affiliation{...}` holds that author’s institution info.

- `\institution{...}` and `\country{...}` go *inside* `\affiliation`. The `acmart` class **requires** a country, which is why every block has `\country{USA}` (that was one of the errors we fixed early on).
- The lone `%` right after `\affiliation{` is a trick: it hides the line break so no stray space sneaks in. A comment “eats” the newline.

Then the abstract and keywords:

```
\begin{abstract}
Large Language Models (LLMs) have resulted in ...
\end{abstract}

\keywords{Large Language Models, Inference Serving, ...}

\maketitle
```

`abstract` is an environment holding the summary paragraph. `\keywords{...}` lists the topic tags. Finally `\maketitle` is the command that actually *draws* the title, authors, and abstract onto the page using all the info above. Nothing appears until `\maketitle` is called.

5. Section headings and cross-references

5.1 Headings

```
\section{Introduction}
\subsection{The LLM Inference Workflow}
\subsubsection{Parameters}
```

These create numbered headings at three levels (1, 1.1, 1.1.1). LaTeX numbers them automatically — if you reorder sections, the numbers update themselves.

5.2 Labels and references

```
\section{Method}
\label{sec:method}
...
... discussed in Section~\ref{sec:method}.
```

`\label{name}` puts an invisible “bookmark” on the thing just before it (a section, figure, table, or equation). `\ref{name}` then prints that thing’s *number* wherever you mention it. So if Method is section 3, `\ref` prints “3”. This is why the report can say “see Section ??” and always get the right number, even after edits. The names like `sec:method`, `fig:transformer`, `tab:datasets`, `eq:optimal` are just labels you choose; the `sec:/fig:/tab:/eq:` prefixes are a common habit to keep them organized.

5.3 Citations and the `~` symbol

```
... office suites~\cite{mehdi2023bing, openai2023chatgpt, spataro2023copilot}.
```

`\cite{key}` prints a reference number like “[5]” and adds that source to the reference list. The key (e.g. mehdi2023bing) must match an entry in your `references.bib` file. You can cite several at once separated by commas.

The `~` (tilde) before `\cite` is a **non-breaking space**: a space that will never be split across a line break. It keeps the word and its citation number together, so you never get “suites” at the end of one line and “[5]” alone at the start of the next. You will see `~` used before `\ref` and `\cite` throughout.

6. Formatting text and special characters

6.1 Font styles

```
\textbf{Dense operations.}    % bold
\emph{compute-bound}         % emphasis (usually italic)
\textsc{NanoFlow}            % SMALL CAPS
\texttt{max-num-tokens}      % typewriter / monospace font
```

6.2 Characters that need special handling

Some characters have special meaning in LaTeX, so to *print* them you escape them with a backslash:

```
50\%           % a literal percent sign (bare % would start a comment!)
72\%           % same
prefill \& decode % a literal & (bare & is a table column separator)
```

Other useful bits you will see:

- `---` becomes a long “em dash” (`—`), used for breaks in a sentence. (`--` is a medium dash for ranges, like 50–72%.)
- Quotes are written with backticks and apostrophes: `‘‘WASTED’’` prints proper curly quotes “WASTED”.
- `\,` inserts a *thin space*, used in numbers with units like 80\,GB or 200\,ms so the number and unit sit close but not touching.
- `\S` prints the section symbol (§), as in `\S\ref{sec:runtime}`.

7. Math and equations

7.1 Inline math

Anything between two dollar signs is typeset as math:

```
Mixtral 8$\times$7B          % the $\times$ prints a multiplication sign
$T_R < 1$ means compute-bound.
$1 \ldots t{-}1$             % \ldots is "...", {-} keeps the minus tidy
```

7.2 Displayed, numbered equations

```
\begin{equation}
  T_{\text{mem}} = \frac{\text{MemSize}}{\text{MemBW}}.
  \label{eq:tmem}
\end{equation}
```

The `equation` environment puts the formula on its own centered line and gives it a number (which you can point at with `\ref{eq:tmem}`). Common pieces inside:

- `\frac{top}{bottom}` makes a fraction.
- `_` is subscript and `^` is superscript (e.g. `\Dmodel^2` is D^2 , `10^{-5}` is 10^{-5}).
- `\approx`, `\le`, `\times`, `\cdot` print $\approx$, $\leq$, $\times$, $\cdot$.
- `\mathit{Dense}` writes a whole word in math italic.

8. Bulleted lists

```
\begin{itemize}
  \item An analytical cost model ...
  \item \nf: an MILP auto-search engine ...
  \item An evaluation showing 1.91$\times$ throughput ...
\end{itemize}
```

`itemize` makes a bulleted list; each bullet starts with `\item`. (A numbered list would use `enumerate` instead.)

9. Figures (inserting images)

```
\begin{figure}[t]
  \centering
  \includegraphics[width=\linewidth]{fig1_transformer.png}
  \caption{Transformer architecture, reproduced from the original paper. ...}
  \label{fig:transformer}
\end{figure}
```

Line by line:

- `\begin{figure}[t]` starts a figure. The `[t]` is a *placement hint* meaning “try to put this at the top of a page.” LaTeX treats figures as “floats” — it slides them to a tidy spot near the text rather than locking them exactly in place.
- `\centering` centers the image horizontally.

- `\includegraphics[width=\linewidth]{...}` inserts the image file and scales it so its width equals `\linewidth` (the full text width). The filename is found via the `figures/` path set earlier.
- `\caption{...}` is the caption printed under the figure; LaTeX adds the “Figure 1” label automatically.
- `\label{fig:transformer}` lets `\ref` refer to it by number.

You will also see `\begin{figure*}` (with a star). The star version spans the *full page width* — used for the wide pipeline diagram.

10. Tables

Tables are the most detailed part of your file. Here is a small version of the pattern:

```
\begin{table}[t]
\centering
\caption{The average and standard deviation of input and output lengths ...}
\label{tab:datasets}
\begin{tabular}{lrr}
\toprule
\textbf{Dataset} & \textbf{Avg. Input (Std)} & \textbf{Avg. Output (Std)} \\
\midrule
Splitwise~\cite{patel2023splitwise} & 1155 (1109) & 211 (163) \\
LMSYS-Chat~\cite{zheng2023lmsys} & 102 (169) & 222 (210) \\
\bottomrule
\end{tabular}
\end{table}
```

Two environments are nested here. The outer `table` behaves like a figure (it floats, has a `\caption` and `\label`). The inner `tabular` holds the actual grid. The key rules:

- `{lrr}` defines the columns: here three columns, one left-aligned and two right-aligned. (c would be centered.) So `{llrrrrrrrrr}` in your big table means ten columns with those alignments.
- Inside a row, `&` separates columns and `\\` ends the row.
- `\toprule`, `\midrule`, `\bottomrule` (from the `booktabs` package) draw the clean horizontal lines at the top, under the header, and at the bottom.

A few extras that appear in your larger tables:

- `\multicolumn{7}{c}{...}` merges 7 columns into one centered cell (used for a header that spans several columns).
- `\multirow{2}{*}{...}` merges a cell down across 2 rows.
- `\cmidrule(lr){2-8}` draws a partial line under only columns 2–8.
- `\setlength{\tabcolsep}{4pt}` reduces the space between columns so a wide table fits.
- `\begin{table*}` (star) again means “span the full page width,” used for the big accelerator table.
- Numbers like 1{,}555 use `{,}` so the comma is spaced as a thousands separator.

11. The algorithm block

```
\begin{algorithm}[t]
\caption{\nf: Auto-search and Runtime Execution}
\label{alg:nanoflow}
\begin{algorithmic}[1]
\Require Model architecture  $\mathcal{M}$ , hardware spec  $\mathcal{H}$ , ...
\Ensure Optimized intra-device pipeline  $\Pi$ , ...
\State  $\mathcal{B}_{\text{dense}}$  \gets \Call{MaxDenseBatch}{...} \Comment{largest batch ...}
\ForAll{k \in \{\text{GEMM}, \text{GEMV}, \text{Network}\}}
  \State ...
\EndFor
\end{algorithmic}
\end{algorithm}
```

Again two nested environments: the outer `algorithm` is a float (like a figure, with caption “Algorithm 1” and a label), and the inner `algorithmic` holds the pseudocode. The `[1]` makes it number the lines. The pseudocode commands come from the `algpseudocode` package:

- `\State` — one ordinary step/line.
- `\Require` / `\Ensure` — the “input” and “output” preconditions at the top.
- `\For`, `\ForAll`, `\EndFor` — a loop and its end.
- `\If`, `\EndIf`, `\While`, `\Repeat`, `\Until` — other control structures.
- `\State $x$ \gets $y$` — the `\gets` prints a left arrow ($\leftarrow$), meaning “assign”.
- `\Call{Name}{args}` — formats a function call.
- `\Comment{...}` — a right-aligned comment on that line.
- `\Statex` — a line with no number (used for the “Phase 0/1/2” headers).

The $\mathcal{M}$ style prints fancy calligraphic capitals used for the model/hardware/workload sets.

12. The bibliography (references)

At the very end of the file:

```
\bibliographystyle{ACM-Reference-Format}
\bibliography{references}
```

- `\bibliography{references}` pulls in your `references.bib` file (the “.bib” is assumed). That .bib file is a database of sources, each with a key (like `kwon2023efficient`), a title, authors, year, etc.
- `\bibliographystyle{ACM-Reference-Format}` chooses *how* the references are formatted — here in the official ACM style your assignment asked for.

- Only sources you actually `\cite` show up in the final list, and they are numbered/ordered automatically. This is why we never had to touch the `.bib` file when removing figures: the citations in the text are what matter.

13. The three fixes we made (in case your teacher asks)

When you first tried to compile, three real errors came up. Here is what they were and the one-line fix for each:

1. “**\Bbbk already defined.**” The `acmart` style and the `amssymb` package both try to define the same symbol. The fix, `\let\Bbbk\relax` placed just before `\usepackage{amssymb}`, temporarily “clears” that symbol so `amssymb` can define it without complaint.
2. “**Undefined control sequence \nf.**” Your `\nf` shortcut uses `\xspace`, but the `xspace` package was not loaded. Adding `\usepackage{xspace}` fixed it.
3. “**No country present for an affiliation.**” The `acmart` class requires every author to have a `\country{...}`. They were empty, so we filled in `\country{USA}`.

One-line summary you can give your teacher: the file uses the required `acmart` document class, loads a handful of standard packages (graphics, math, tables, algorithms), defines a couple of custom shortcuts, builds the title block, and then writes the content using sections, figures, tables, equations, an algorithm, and citations — all of which LaTeX numbers and cross-references automatically.